

ECC Pipeline — Non-Technical Overview

What it does

The ECC pipeline takes a list of North Carolina business entities — companies, nonprofits, fire departments, churches — and finds a verified, deliverable email address for each one. It does this automatically, without human intervention, by combining three techniques:

1. It looks up the business's website by guessing the domain name (e.g. acme.com from "Acme Corp").
 2. It searches Google for contact information if the domain guess fails.
 3. It checks whether an email address actually works without sending an email, using specialized verification services.
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What comes out

Three files per run:

File	What it contains
valid_emails.csv	One row per confirmed email address
results.json	Run summary: how many succeeded, failed, and what it cost
pipeline.db	Complete database of every record processed (internal use)

Reading the CSV

Each row in valid_emails.csv has nine columns:

Column	Plain English
unique_id	Internal ID tying this email back to the original filing
business_name	The legal business name
agent_name	The registered agent or officer on file
state	Two-letter state code (always NC for this dataset)
email	The confirmed deliverable email
zuhal_status	The raw result from the verification service

confidence_tier	How reliable this email is: high, medium, or low
discovery_method	How the email was found: DNS lookup, Google search, or already in the data
validation_method	Which service confirmed it works: Zuhal, Microsoft, or bbops

What "confidence" means

Tier	Meaning	Typical example
High	Email is on the business's own domain and matches a known contact pattern	info@carolinasolventsinc.com
Medium	Email is domain-matched but the server accepts all addresses, or it is a generic prefix	info@asheville.com
Low	Email was verified but is a personal address (Gmail, AOL) or the domain could not be confirmed	glenwoodchief441@gmail.com

Rule of thumb: High and medium emails are reliable for outreach. Low emails are real but may not reach the decision-maker.

How email addresses are found

Step 1 — DNS lookup

The pipeline guesses the business's domain by stripping legal words ("Inc", "LLC", "Foundation") from the name and checking if that domain has a mail server. Example: "Acme Corporation" tries acme.com, acme.net, acme.org.

Step 2 — Google search

If DNS finds nothing, the pipeline searches Google for the business name with contact and email keywords and extracts any email addresses found in the results.

Step 3 — Pattern generation

Once a domain is confirmed, the pipeline generates likely email patterns: info@domain.com, contact@domain.com, first.last@domain.com, etc.

How email addresses are verified

Three services are used, in order of preference:

Service	Cost	How it works
bbops.io	Per contract	Sends a simulated SMTP connection to check if the mailbox exists — no actual email sent
MS Probe	Free	For businesses on Microsoft 365, Microsoft's own login system confirms if the user account exists
Zuhal	~\$0.0005 per check	Paid SMTP verification service; last resort

The pipeline tries free verification first and only falls back to paid when necessary.

Why some businesses don't get an email

About 80% of the NC nonprofit and volunteer fire department dataset has no discoverable web presence — no domain registered, no Google listing. The pipeline marks these as Discovery Failed. This reflects the source data quality, not a pipeline defect.

For comparison, commercial datasets (LLCs, corporations) typically yield a 50–70% discovery rate.

Cost

Scale	Estimated cost
500 records (tested)	\$0.58
10,000 records	~\$12
100,000 records	~\$120
300,000 records (full NC dataset)	~\$350

A cost ceiling can be set. The pipeline stops spending once the limit is reached.

Live Run — 500 Records

Date: 2026-04-30 Input: NC Secretary of State filing records Mode: Live (real API calls)

Summary

Metric	Value
Records processed	500
Discovery failed	401
Validation failed	55
Validated emails	44
Overall yield	8.8%

Confidence

Tier	Count
High	6
Medium	14
Low	24

Verdict

Verdict	Count
Valid	35
Accept-all	7
MS Valid	2

Discovery Method

Method	Count
DNS	15
Serper	1
Not recorded	28

Validation Method

Method	Count
Zuhal	42

MS Probe	2
bbops	0

Cost

Metric	Value
Total cost	\$0.58
Cost per record	\$0.00116
Cost per validated email	\$0.0131